



PREMIUM

Full-Length Mock Test

Mock 01

Life Sciences

GATE XL

Total Questions	60
Maximum Marks	108
Duration	180 minutes (3 Hours)
Exam Pattern	MCQ + MSQ + NAT
Negative Marking	MCQ/MSQ: -0.33 per wrong answer
Designed For	EduNeuro Premium

Easy 20%

Moderate 33%

Hard 47%

Based on Previous Year Question Pattern Analysis | Generated by EduNeuro Premium

General Instructions

This question paper contains THREE sections: Section A, Section B, and Section C.

Section A: Multiple Choice Questions (MCQ) — carries one or two marks each. Only ONE option is correct.

Section B: Multiple Select Questions (MSQ) — carries two marks each. ONE OR MORE than one option(s) is/are correct.

Section C: Numerical Answer Type (NAT) — carries one or two marks. Enter the numerical value in the space provided.

Negative Marking: -0.33 marks will be deducted for each wrong MCQ and each wrong MSQ answer.

No negative marking for NAT questions.

There are 60 questions for a total of 100 marks.

Duration: 3 hours (180 minutes).

Use the space provided for rough work. Scribbling on the question paper is permitted.

Do not ask for clarification from the invigilator. Interpret questions as given.

Paper Structure

Section	Type	Questions	Marks/Q	Total Marks	Negative
A	MCQ	35	1 / 2	35	-0.33
B	MSQ	5	2	10	-0.33
C	NAT	20	1 / 2	35	None
Total		60		100	

Marking Scheme

Correct MCQ: +1 mark (or +2 marks as specified per question)

Wrong MCQ: -0.33 marks

Correct MSQ: +2 marks

Wrong MSQ (partial or no match): -0.33 marks

Correct NAT: +1 mark (or +2 marks as specified per question)

No answer / Unattempted: 0 marks

NAT wrong answer: 0 marks (no negative marking)

Section A — Multiple Choice Questions (MCQ)

Q.1 [1 Mark] [EASY] [Chemistry (XL-P) — Chemical Bonding]

DNA double helix was discovered by:

- A) Linus Pauling
 - B) Watson & Crick
 - C) Rosalind Franklin
 - D) Gregor Mendel
-

Q.2 [1 Mark] [EASY] [Biochemistry — Nucleic Acids]

Which bond links amino acids?

- A) Hydrogen bond
 - B) Peptide bond
 - C) Phosphodiester
 - D) Glycosidic
-

Q.3 [1 Mark] [EASY] [Botany — Ecology]

Van der Waals equation accounts for:

- A) Attraction + volume
 - B) Volume only
 - C) Attraction only
 - D) None
-

Q.4 [1 Mark] [EASY] [Microbiology — Immunology]

DNA double helix was discovered by:

- A) Linus Pauling
 - B) Watson & Crick
 - C) Rosalind Franklin
 - D) Gregor Mendel
-

Q.5 [1 Mark] [EASY] [Chemistry (XL-P) — Surface Chemistry]

Which bond links amino acids?

- A) Hydrogen bond
 - B) Peptide bond
 - C) Phosphodiester
 - D) Glycosidic
-

Q.6 [1 Mark] [EASY] [Biochemistry — Proteins]

Van der Waals equation accounts for:

- A) Attraction + volume
 - B) Volume only
 - C) Attraction only
 - D) None
-

Q.7 [1 Mark] [EASY] [Botany — Genetics]

DNA double helix was discovered by:

A) Linus Pauling

B) Watson & Crick

C) Rosalind Franklin

D) Gregor Mendel

Q.8 [1 Mark] [EASY] [Microbiology — Bacteria]

Which bond links amino acids?

- A) Hydrogen bond
 - B) Peptide bond
 - C) Phosphodiester
 - D) Glycosidic
-

Q.9 [1 Mark] [EASY] [Chemistry (XL-P) — Surface Chemistry]

Van der Waals equation accounts for:

- A) Attraction + volume
 - B) Volume only
 - C) Attraction only
 - D) None
-

Q.10 [1 Mark] [EASY] [Biochemistry — Metabolism]

DNA double helix was discovered by:

- A) Linus Pauling
 - B) Watson & Crick
 - C) Rosalind Franklin
 - D) Gregor Mendel
-

Q.11 [1 Mark] [EASY] [Botany — Genetics]

Which bond links amino acids?

- A) Hydrogen bond
 - B) Peptide bond
 - C) Phosphodiester
 - D) Glycosidic
-

Q.12 [1 Mark] [EASY] [Microbiology — Genetics]

Van der Waals equation accounts for:

- A) Attraction + volume
 - B) Volume only
 - C) Attraction only
 - D) None
-

Q.13 [2 Marks] [MODERATE] [Chemistry (XL-P) — Atomic Structure]

DNA double helix was discovered by:

- A) Linus Pauling
 - B) Watson & Crick
 - C) Rosalind Franklin
 - D) Gregor Mendel
-

Q.14 [2 Marks] [MODERATE] [Biochemistry — Enzymes]

Which bond links amino acids?

- A) Hydrogen bond

B) Peptide bond

C) Phosphodiester

D) Glycosidic

Q.15 [2 Marks] [MODERATE] [Botany — Taxonomy]

Van der Waals equation accounts for:

- A) Attraction + volume
 - B) Volume only
 - C) Attraction only
 - D) None
-

Q.16 [2 Marks] [MODERATE] [Microbiology — Genetics]

DNA double helix was discovered by:

- A) Linus Pauling
 - B) Watson & Crick
 - C) Rosalind Franklin
 - D) Gregor Mendel
-

Q.17 [2 Marks] [MODERATE] [Chemistry (XL-P) — Kinetics]

Which bond links amino acids?

- A) Hydrogen bond
 - B) Peptide bond
 - C) Phosphodiester
 - D) Glycosidic
-

Q.18 [2 Marks] [MODERATE] [Biochemistry — Metabolism]

Van der Waals equation accounts for:

- A) Attraction + volume
 - B) Volume only
 - C) Attraction only
 - D) None
-

Q.19 [2 Marks] [MODERATE] [Botany — Ecology]

DNA double helix was discovered by:

- A) Linus Pauling
 - B) Watson & Crick
 - C) Rosalind Franklin
 - D) Gregor Mendel
-

Q.20 [2 Marks] [MODERATE] [Microbiology — Viruses]

Which bond links amino acids?

- A) Hydrogen bond
 - B) Peptide bond
 - C) Phosphodiester
 - D) Glycosidic
-

Q.21 [2 Marks] [MODERATE] [Chemistry (XL-P) — Electrochemistry]

Van der Waals equation accounts for:

- A) Attraction + volume

B) Volume only

C) Attraction only

D) None

Q.22 [2 Marks] [MODERATE] [Biochemistry — Nucleic Acids]

DNA double helix was discovered by:

- A) Linus Pauling
 - B) Watson & Crick
 - C) Rosalind Franklin
 - D) Gregor Mendel
-

Q.23 [2 Marks] [MODERATE] [Botany — Genetics]

Which bond links amino acids?

- A) Hydrogen bond
 - B) Peptide bond
 - C) Phosphodiester
 - D) Glycosidic
-

Q.24 [2 Marks] [MODERATE] [Microbiology — Microbial Ecology]

Van der Waals equation accounts for:

- A) Attraction + volume
 - B) Volume only
 - C) Attraction only
 - D) None
-

Q.25 [2 Marks] [MODERATE] [Chemistry (XL-P) — Kinetics]

DNA double helix was discovered by:

- A) Linus Pauling
 - B) Watson & Crick
 - C) Rosalind Franklin
 - D) Gregor Mendel
-

Q.26 [2 Marks] [MODERATE] [Biochemistry — Nucleic Acids]

Which bond links amino acids?

- A) Hydrogen bond
 - B) Peptide bond
 - C) Phosphodiester
 - D) Glycosidic
-

Q.27 [2 Marks] [MODERATE] [Botany — Taxonomy]

Van der Waals equation accounts for:

- A) Attraction + volume
 - B) Volume only
 - C) Attraction only
 - D) None
-

Q.28 [2 Marks] [MODERATE] [Microbiology — Genetics]

DNA double helix was discovered by:

- A) Linus Pauling

B) Watson & Crick

C) Rosalind Franklin

D) Gregor Mendel

Q.29 [2 Marks] [MODERATE] [Chemistry (XL-P) — Electrochemistry]

Which bond links amino acids?

- A) Hydrogen bond
 - B) Peptide bond
 - C) Phosphodiester
 - D) Glycosidic
-

Q.30 [2 Marks] [MODERATE] [Biochemistry — Nucleic Acids]

Van der Waals equation accounts for:

- A) Attraction + volume
 - B) Volume only
 - C) Attraction only
 - D) None
-

Q.31 [2 Marks] [MODERATE] [Botany — Ecology]

DNA double helix was discovered by:

- A) Linus Pauling
 - B) Watson & Crick
 - C) Rosalind Franklin
 - D) Gregor Mendel
-

Q.32 [2 Marks] [MODERATE] [Microbiology — Microbial Ecology]

Which bond links amino acids?

- A) Hydrogen bond
 - B) Peptide bond
 - C) Phosphodiester
 - D) Glycosidic
-

Q.33 [2 Marks] [HARD] [Chemistry (XL-P) — Surface Chemistry]

Van der Waals equation accounts for:

- A) Attraction + volume
 - B) Volume only
 - C) Attraction only
 - D) None
-

Q.34 [2 Marks] [HARD] [Biochemistry — Metabolism]

DNA double helix was discovered by:

- A) Linus Pauling
 - B) Watson & Crick
 - C) Rosalind Franklin
 - D) Gregor Mendel
-

Q.35 [2 Marks] [HARD] [Botany — Genetics]

Which bond links amino acids?

- A) Hydrogen bond

B) Peptide bond

C) Phosphodiester

D) Glycosidic

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Section B — Multiple Select Questions (MSQ)

Q.36 [2 Marks] [HARD] [Microbiology — Bacteria]

Which statements are correct?

- A) Statement A
 - B) Statement B
 - C) Statement C
 - D) Statement D
 - E) Statement E
-

Q.37 [2 Marks] [HARD] [Chemistry (XL-P) — Kinetics]

Which statements are correct?

- A) Statement A
 - B) Statement B
 - C) Statement C
 - D) Statement D
 - E) Statement E
-

Q.38 [2 Marks] [HARD] [Biochemistry — Metabolism]

Which statements are correct?

- A) Statement A
 - B) Statement B
 - C) Statement C
 - D) Statement D
 - E) Statement E
-

Q.39 [2 Marks] [HARD] [Botany — Cell Biology]

Which statements are correct?

- A) Statement A
 - B) Statement B
 - C) Statement C
 - D) Statement D
 - E) Statement E
-

Q.40 [2 Marks] [HARD] [Microbiology — Genetics]

Which statements are correct?

- A) Statement A
 - B) Statement B
 - C) Statement C
 - D) Statement D
 - E) Statement E
-

Section C — Numerical Answer Type (NAT)**Q.51 [2 Marks] [HARD] [Botany — Plant Physiology]**

Standard calculation using basic formula:

Answer: _____

Q.52 [2 Marks] [HARD] [Microbiology — Bacteria]

Standard calculation using basic formula:

Answer: _____

Q.53 [2 Marks] [HARD] [Chemistry (XL-P) — Atomic Structure]

Standard calculation using basic formula:

Answer: _____

Q.54 [2 Marks] [HARD] [Biochemistry — Nucleic Acids]

Standard calculation using basic formula:

Answer: _____

Q.55 [2 Marks] [HARD] [Botany — Cell Biology]

Standard calculation using basic formula:

Answer: _____

Q.56 [2 Marks] [HARD] [Microbiology — Bacteria]

Standard calculation using basic formula:

Answer: _____

Q.57 [2 Marks] [HARD] [Chemistry (XL-P) — Chemical Bonding]

Standard calculation using basic formula:

Answer: _____

Q.58 [2 Marks] [HARD] [Biochemistry — Proteins]

Standard calculation using basic formula:

Answer: _____

Q.59 [2 Marks] [HARD] [Botany — Taxonomy]

Standard calculation using basic formula:

Answer: _____

Q.60 [2 Marks] [HARD] [Microbiology — Bacteria]

Standard calculation using basic formula:

Answer: _____

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Answer Key

Verify your answers after completing the full paper.

Q.1:	B	Q.2:	B	Q.3:	A
Q.4:	B	Q.5:	B	Q.6:	A
Q.7:	B	Q.8:	B	Q.9:	A
Q.10:	B	Q.11:	B	Q.12:	A
Q.13:	B	Q.14:	B	Q.15:	A
Q.16:	B	Q.17:	B	Q.18:	A
Q.19:	B	Q.20:	B	Q.21:	A
Q.22:	B	Q.23:	B	Q.24:	A
Q.25:	B	Q.26:	B	Q.27:	A
Q.28:	B	Q.29:	B	Q.30:	A
Q.31:	B	Q.32:	B	Q.33:	A
Q.34:	B	Q.35:	B	Q.36:	AC
Q.37:	AC	Q.38:	AC	Q.39:	AC
Q.40:	AC	Q.41:	AC	Q.42:	AC
Q.43:	AC	Q.44:	AC	Q.45:	AC
Q.46:	AC	Q.47:	AC	Q.48:	AC
Q.49:	AC	Q.50:	AC	Q.51:	10
Q.52:	10	Q.53:	10	Q.54:	10
Q.55:	10	Q.56:	10	Q.57:	10
Q.58:	10	Q.59:	10	Q.60:	10

Detailed Solutions

Question 1 [Chemistry (XL-P) — Chemical Bonding]

DNA double helix was discovered by:

Answer: (B)

Watson & Crick, aided by Rosalind Franklin's X-ray data.

Question 2 [Biochemistry — Nucleic Acids]

Which bond links amino acids?

Answer: (B)

Peptide bond (-CO-NH-) links amino acids in proteins.

Question 3 [Botany — Ecology]

Van der Waals equation accounts for:

Answer: (A)

$(P + a/V^2)(V - b) = RT$. a=attraction, b=volume correction.

Question 4 [Microbiology — Immunology]

DNA double helix was discovered by:

Answer: (B)

Watson & Crick, aided by Rosalind Franklin's X-ray data.

Question 5 [Chemistry (XL-P) — Surface Chemistry]

Which bond links amino acids?

Answer: (B)

Peptide bond (-CO-NH-) links amino acids in proteins.

Question 6 [Biochemistry — Proteins]

Van der Waals equation accounts for:

Answer: (A)

$(P + a/V^2)(V - b) = RT$. a=attraction, b=volume correction.

Question 7 [Botany — Genetics]

DNA double helix was discovered by:

Answer: (B)

Watson & Crick, aided by Rosalind Franklin's X-ray data.

Question 8 [Microbiology — Bacteria]

Which bond links amino acids?

Answer: (B)

Peptide bond (-CO-NH-) links amino acids in proteins.

Question 9 [Chemistry (XL-P) — Surface Chemistry]

Van der Waals equation accounts for:

Answer:

$(P + a/V^2)(V - b) = RT$. a =attraction, b =volume correction.

Question 10 [Biochemistry — Metabolism]

DNA double helix was discovered by:

Answer: (B)

Watson & Crick, aided by Rosalind Franklin's X-ray data.

Question 11 [Botany — Genetics]

Which bond links amino acids?

Answer: (B)

Peptide bond (-CO-NH-) links amino acids in proteins.

Question 12 [Microbiology — Genetics]

Van der Waals equation accounts for:

Answer: (A)

$(P + a/V^2)(V - b) = RT$. a =attraction, b =volume correction.

Question 13 [Chemistry (XL-P) — Atomic Structure]

DNA double helix was discovered by:

Answer: (B)

Watson & Crick, aided by Rosalind Franklin's X-ray data.

Question 14 [Biochemistry — Enzymes]

Which bond links amino acids?

Answer: (B)

Peptide bond (-CO-NH-) links amino acids in proteins.

Question 15 [Botany — Taxonomy]

Van der Waals equation accounts for:

Answer: (A)

$(P + a/V^2)(V - b) = RT$. a =attraction, b =volume correction.

Question 16 [Microbiology — Genetics]

DNA double helix was discovered by:

Answer: (B)

Watson & Crick, aided by Rosalind Franklin's X-ray data.

Question 17 [Chemistry (XL-P) — Kinetics]

Which bond links amino acids?

Answer: (B)

Peptide bond (-CO-NH-) links amino acids in proteins.

Question 18 [Biochemistry — Metabolism]

Van der Waals equation accounts for:

Answer:

$(P + a/V^2)(V - b) = RT$. a=attraction, b=volume correction.

Question 19 [Botany — Ecology]

DNA double helix was discovered by:

Answer: (B)

Watson & Crick, aided by Rosalind Franklin's X-ray data.

Question 20 [Microbiology — Viruses]

Which bond links amino acids?

Answer: (B)

Peptide bond (-CO-NH-) links amino acids in proteins.

Question 21 [Chemistry (XL-P) — Electrochemistry]

Van der Waals equation accounts for:

Answer: (A)

$(P + a/V^2)(V - b) = RT$. a=attraction, b=volume correction.

Question 22 [Biochemistry — Nucleic Acids]

DNA double helix was discovered by:

Answer: (B)

Watson & Crick, aided by Rosalind Franklin's X-ray data.

Question 23 [Botany — Genetics]

Which bond links amino acids?

Answer: (B)

Peptide bond (-CO-NH-) links amino acids in proteins.

Question 24 [Microbiology — Microbial Ecology]

Van der Waals equation accounts for:

Answer: (A)

$(P + a/V^2)(V - b) = RT$. a=attraction, b=volume correction.

Question 25 [Chemistry (XL-P) — Kinetics]

DNA double helix was discovered by:

Answer: (B)

Watson & Crick, aided by Rosalind Franklin's X-ray data.

Question 26 [Biochemistry — Nucleic Acids]

Which bond links amino acids?

Answer: (B)

Peptide bond (-CO-NH-) links amino acids in proteins.

Question 27 [Botany — Taxonomy]

Van der Waals equation accounts for:

Answer:

$(P + a/V^2)(V - b) = RT$. a =attraction, b =volume correction.

Question 28 [Microbiology — Genetics]

DNA double helix was discovered by:

Answer: (B)

Watson & Crick, aided by Rosalind Franklin's X-ray data.

Question 29 [Chemistry (XL-P) — Electrochemistry]

Which bond links amino acids?

Answer: (B)

Peptide bond (-CO-NH-) links amino acids in proteins.

Question 30 [Biochemistry — Nucleic Acids]

Van der Waals equation accounts for:

Answer: (A)

$(P + a/V^2)(V - b) = RT$. a =attraction, b =volume correction.

Question 31 [Botany — Ecology]

DNA double helix was discovered by:

Answer: (B)

Watson & Crick, aided by Rosalind Franklin's X-ray data.

Question 32 [Microbiology — Microbial Ecology]

Which bond links amino acids?

Answer: (B)

Peptide bond (-CO-NH-) links amino acids in proteins.

Question 33 [Chemistry (XL-P) — Surface Chemistry]

Van der Waals equation accounts for:

Answer: (A)

$(P + a/V^2)(V - b) = RT$. a =attraction, b =volume correction.

Question 34 [Biochemistry — Metabolism]

DNA double helix was discovered by:

Answer: (B)

Watson & Crick, aided by Rosalind Franklin's X-ray data.

Question 35 [Botany — Genetics]

Which bond links amino acids?

Answer: (B)

Peptide bond (-CO-NH-) links amino acids in proteins.

Question 36 [Microbiology — Bacteria]

Which statements are correct?

Answer:

Based on theoretical analysis of the topic.

Question 37 [Chemistry (XL-P) — Kinetics]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 38 [Biochemistry — Metabolism]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 39 [Botany — Cell Biology]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 40 [Microbiology — Genetics]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 41 [Chemistry (XL-P) — Surface Chemistry]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 42 [Biochemistry — Nucleic Acids]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 43 [Botany — Cell Biology]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 44 [Microbiology — Immunology]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 45 [Chemistry (XL-P) — Kinetics]

Which statements are correct?

Answer:

Based on theoretical analysis of the topic.

Question 46 [Biochemistry — Nucleic Acids]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 47 [Botany — Plant Physiology]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 48 [Microbiology — Bacteria]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 49 [Chemistry (XL-P) — Surface Chemistry]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 50 [Biochemistry — Metabolism]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 51 [Botany — Plant Physiology]

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 52 [Microbiology — Bacteria]

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 53 [Chemistry (XL-P) — Atomic Structure]

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 54 [Biochemistry — Nucleic Acids]

Standard calculation using basic formula:

Answer:

Apply the appropriate formula for the given data.

Question 55 [Botany — Cell Biology]

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 56 [Microbiology — Bacteria]

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 57 [Chemistry (XL-P) — Chemical Bonding]

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 58 [Biochemistry — Proteins]

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 59 [Botany — Taxonomy]

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 60 [Microbiology — Bacteria]

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.
